

7.2 How can materials and processes be used to make final prototypes?

- a. Understand methods of joining similar and dissimilar materials within products to fulfill the following functions:
- permanently joining materials to include constructional joints
 - temporarily/semi-permanently joining materials
 - adhesion and heat
 - using standard components and fixings.



- b. Demonstrate an understanding of a variety of processes, tools and machinery used to accurately manufacture final prototypes in the workshop made from wood, metal and polymers, including:
- wasting techniques, such as drilling, sawing, shaping and abrading
 - moulding methods, such as thermoforming and vacuum forming
 - milling metals and turning woods
 - casting of metals such as lost wax casting, sand casting, low temperature and resin casting
 - forming and lamination
 - bending, rolling and forming sheet material.



- c. Understand how digital technology, including the use of computer-aided design (CAD) and computer-aided manufacture (CAM) can be used in the making of final prototypes.

- d. Understand how the design of templates, jigs, formers and moulds ensure quality and accuracy when making a final prototype.



- e. Understand how the available forms, costs and working properties of materials contribute to the decisions about suitability of materials when developing and manufacturing products.



